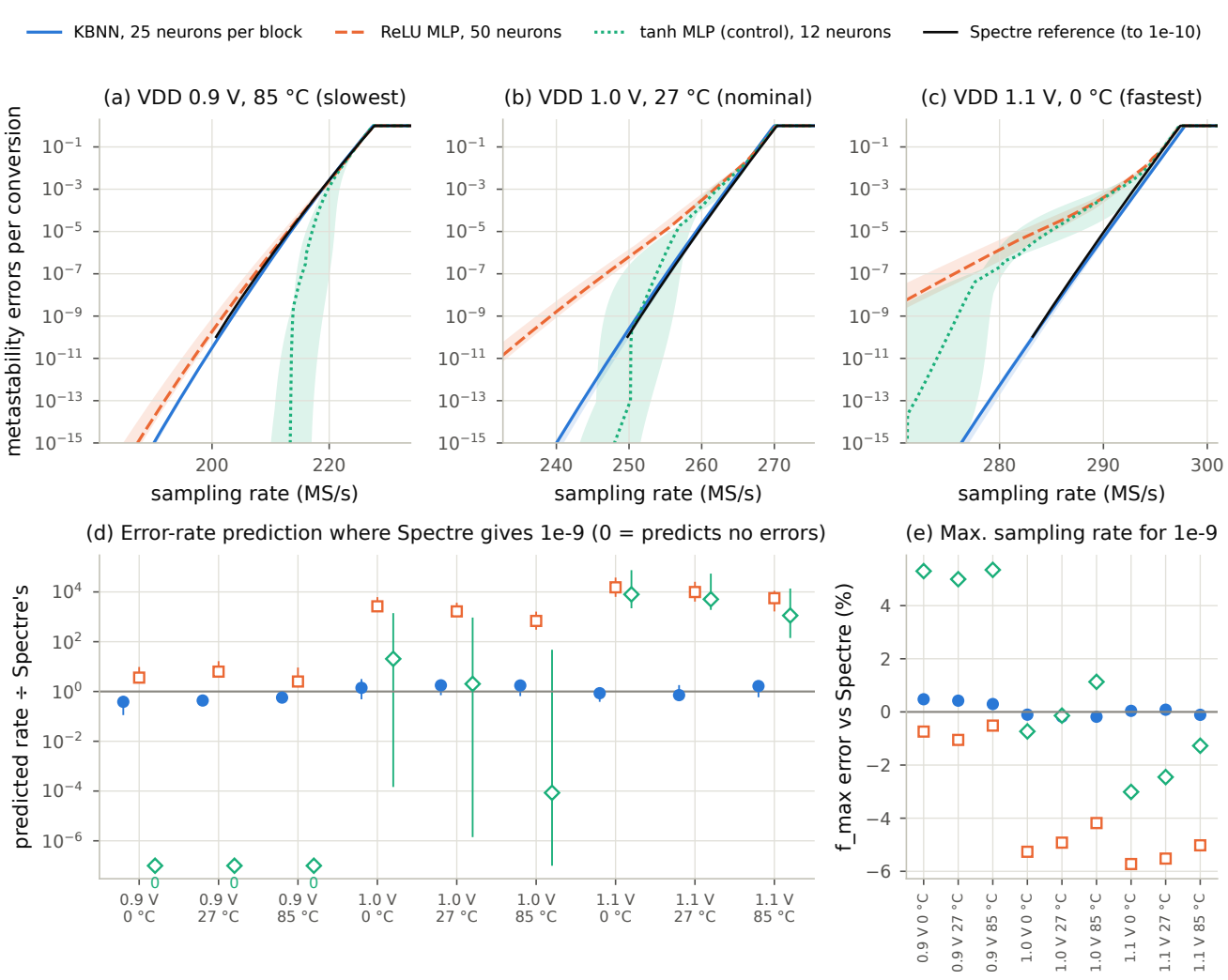

KBNN, 25 neurons per block
ReLU MLP, 50 neurons
tanh MLP (control), 12 neurons
Spectre reference (to 1e-10)
(a) VDD 0.9 V, 85 °C (slowest)
(b) VDD 1.0 V, 27 °C (nominal)
(c) VDD 1.1 V, 0 °C (fastest)
metastability errors per conversion
sampling rate (MS/s)
(d) Error-rate prediction where Spectre gives 1e-9 (0 = predicts no errors)
predicted rate ÷ Spectre's
0.9 V 0 °C
0.9 V 27 °C
0.9 V 85 °C
1.0 V 0 °C
1.0 V 27 °C
1.0 V 85 °C
1.1 V 0 °C
1.1 V 27 °C
1.1 V 85 °C
(e) Max. sampling rate for 1e-9
f_max error vs Spectre (%)